Building surface model

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Content

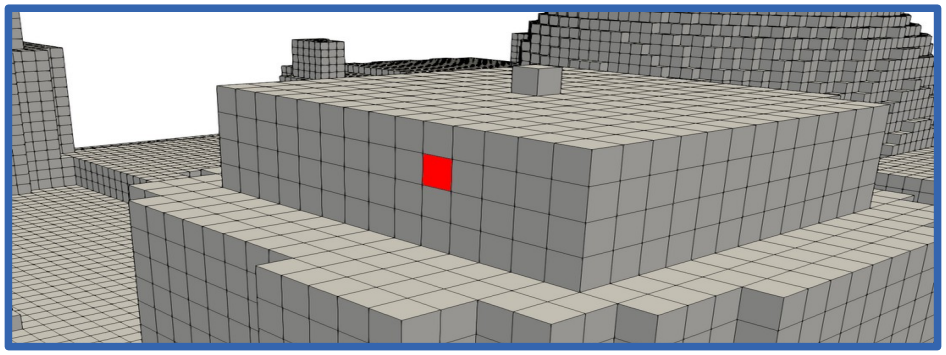
- General concept of the building surface model
- Differences to land surface model
- Usage
- Examples
- Indoor climate model: Outline

General concept

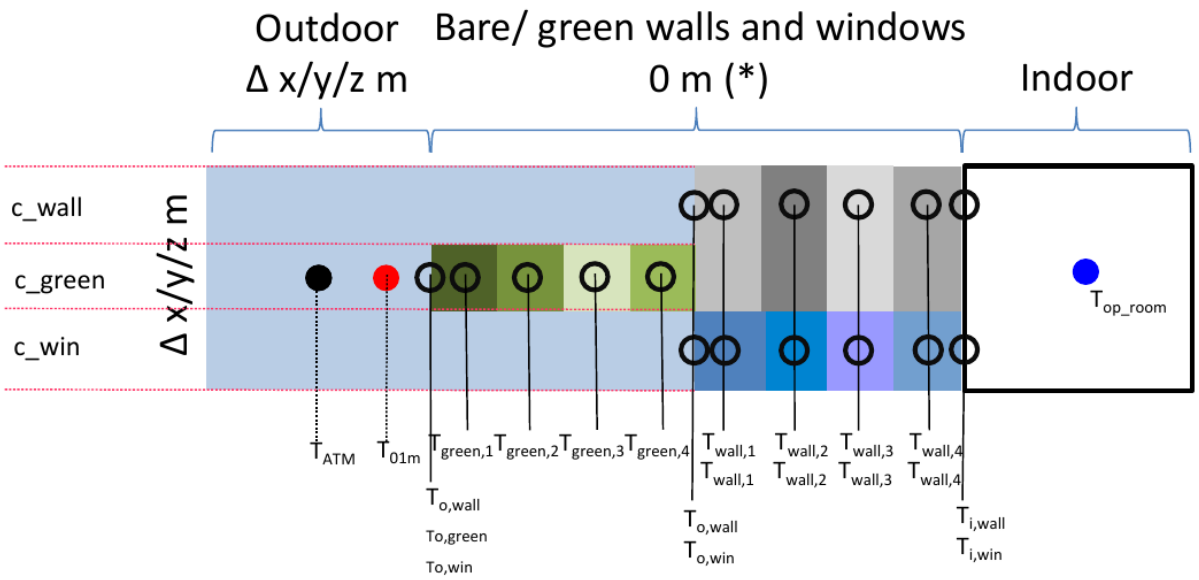
- Representation of buildings as obstacles
- Commonly done in CFD models, but
  - No thermodynamics, or
  - Prescribed surface temperature / fluxes at building surfaces.
  - Thermodynamic interactions are often ignored
- If interactions between atmosphere and buildings are important:
  - Building surface model (BSM)
- Treatment similar as in Land surface model (LSM)
  (1D, building surfaces on Cartesian grid)
- Coupled to optional indoor climate model (ICM)
- Terms BSM and USM (urban surface model, deprecated) used synonymously

General concept

- Energy balance solver for T_0
- Tile approach:
  - Wall fraction c_wall
  - Window fraction c_win
  - Green fraction c_green



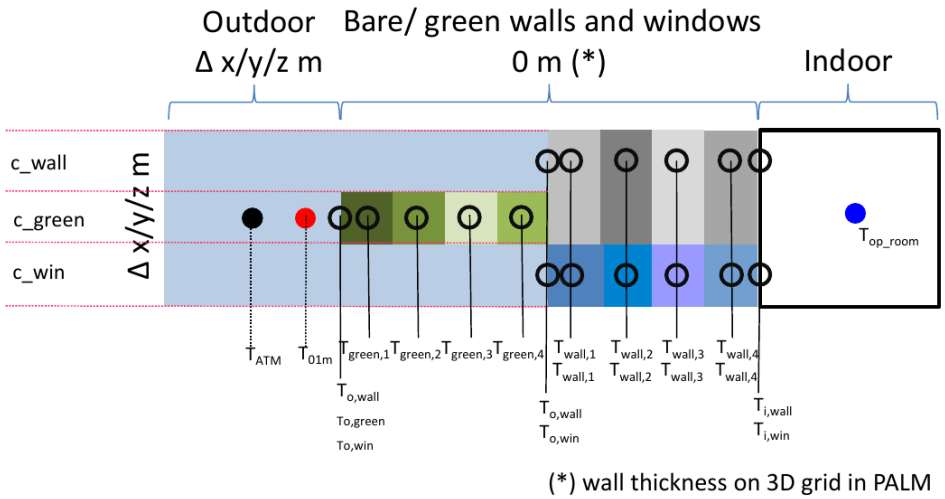
- 3-layer wall model
- Boundary condition:
  - ICM: T_{op_room}
  - no ICM: $T_{wall,4}$



(*) wall thickness on 3D grid in PALM

General concept

- Physical properties of surfaces, walls, windows, green elements are stored in a database
- Insulation of windows is characterized by the U-value (in a single layer)
- Absorption of shortwave radiation inside window layers follows logarithmic function
- Absorbed heat by windows is taken into account
- Green elements similar to LSM, but extraction of water from deeper soil layers by plants is neglected (substrate is saturated for now)



- For details, see Resler et al. (2017, Geosci. Model Dev. , Vol 10)

Building database

- Physical properties of surfaces, walls, windows, green elements are stored in a database
- Classification:

building_type	Description (Usage, Year of construction)
0	User-defined
1	Residential, < 1950
2	Residential, 1950 - 2000
3	Residential, > 2000
4	Office, < 1950
5	Office, 1950 - 2000
6	Office, > 2000

Building database

- Each building_type in the database sets > 100 parameters automatically
- Each parameter is set for ground floor level, above ground floor level and roof separately
- Most important parameters:

  - wall, window, green fraction on surface
  - leaf area index of green fraction
  - roughness length (heat, moisture, momentum)
  - window transmissivity
  - surface albedo (type)
  - surface emissivity
  - heat capacity for each layer
  - thermal conductivity for each layer
  - depth of each wall/roof layer

Building database

- Parameters for the indoor model

  - target temperatures for summer and winter
  - airflow parameters (ventilation)
  - maximum heating capacity
  - maximum cooling capacity
  - internal heat gains
  - storey height
  - g- and u-values
  - ...

Building database

- Each building_type in the database sets all parameters automatically
- Most important parameters can be overwritten in the static driver → e.g. green fraction, surface albedo

Technical details

- Calculation of surface resistances for vertical surfaces (LSM and BSM)

Horizontal surfaces:
Monin-Obukhov Similarity - theory

$$H = -\rho\, c_p\, \frac{1}{r_a}(\theta_1 - \theta_0)$$
$$= \rho\, c_p\, \overline{w'\theta'}_0$$
$$= -\rho\, c_p\, u_*\, \theta_*$$

$$u_*\, \theta_* = \frac{1}{r_a}(\theta_1 - \theta_0)$$

$$\boxed{\rightarrow r_a = \frac{\theta_1 - \theta_0}{u_*\, \theta_*}}$$

Vertical surfaces :
Krayenhoff & Voogt (2007) - empirical

$$H = \left(\frac{z_0}{z_{0,\mathrm{concrete}}}(11.8 - 4.2U) - 4.0\right)(\theta_1 - \theta_0)$$

$$\boxed{\rightarrow r_a = \frac{\rho c_p}{\left(\frac{z_0}{z_{0,\mathrm{concrete}}}(11.8 - 4.2U) - 4.0\right)}}$$

Namelist input parameters

- If NAMELIST is set BSM, is activated
- If no ICM is used:
  set inner temperatures as boundary conditions
- Attention: there are some deprecated parameters that should not be used (especially when a static driver provides all information)
- See also
  https://docs.palm-model.org/25.04/Reference/LES_Model/Namelists/#urban-surface-parameters

```
&urban_surface_parameters

   wall_inner_temperature = 295.0,
   window_inner_temperature = 295.0,
   roof_inner_temperature = 295.0,

 /
```

Input from static driver

- External NetCDF file provided by the user in INPUT folder

- Example: building types are stored in variable

  `building_type(y,x)`

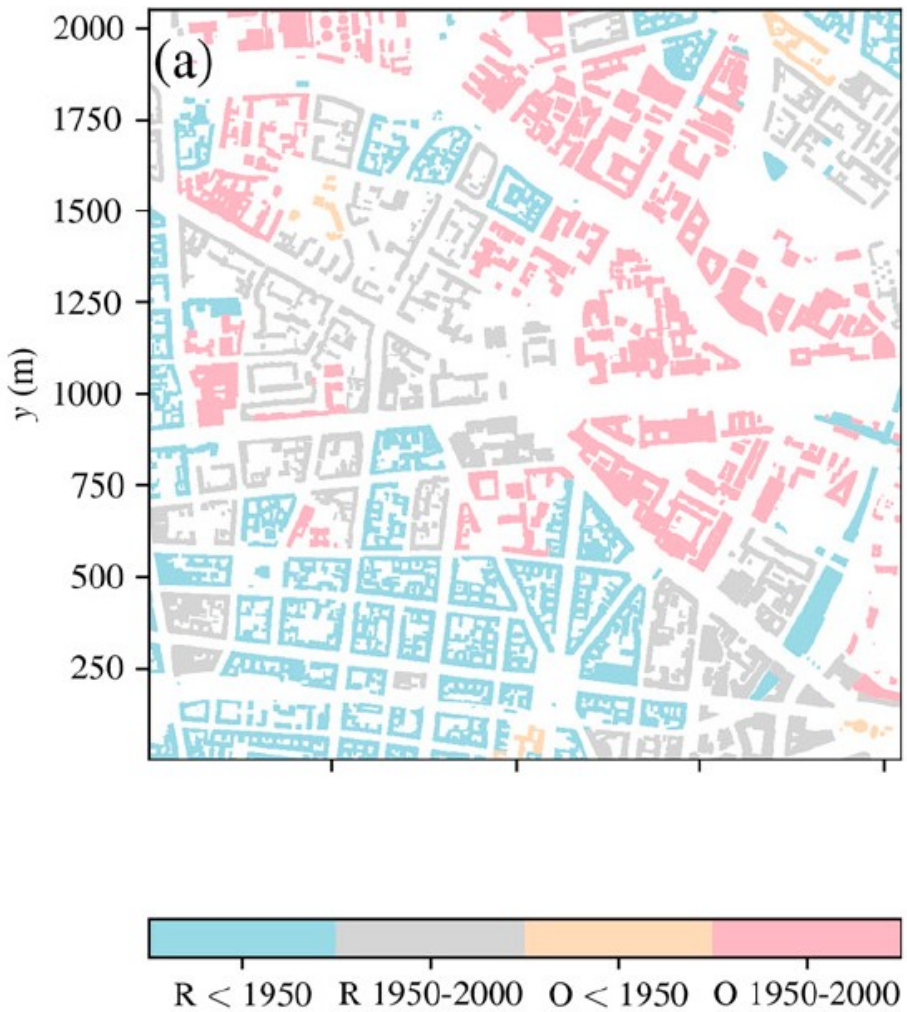
  → 2D map, values 1-6

- Default parameters can be overwritt e.g.

  `building_heat_capacity(type,laye`

- Individiual surface properties can be overwritten, e.g.

  `building_surface_pars(npars, ns)`



Input from static driver

- Documentation for all available parameters:
- https://docs.palm-model.org/25.04/Reference/LES_Model/IO-Files/Drivers/static/

- See lecture about static drivers later today!

Other requirements

- In order to use BSM (or LSM in complex terrain), a special scheme for urban radiative transfer is needed (RTM)
- Since the BSM/USM only handles buildings, the LSM ist usually required to treat non-building surfaces (pavements, vegetation, soil, water)
- Indoor climate model can be switched on

  → Useful for longer simulation periods, where indoor temperature might change significantly

  → Useful for situations where waste heat due to building heating/cooling plays a role

Output variables (note: „usm“-notation still active!)

```
&runtime_parameters

   data_output = ...,
           'usm_t_wall_north',
           'usm_t_window_south,
           'usm_t_green_west',
           'usm_swc_east',

           'usm_surfz',
           'usm_surfcat',
           'usm_surfwintrans',
           'usm_wshf',
           'usm_qsws', 'usm_qsws_veg', 'usm_qsws_liq',
           'usm_wghf', 'usm_wghf_window', 'usm_wghf_green',
           'usm_ighf', 'usm_ighf_window', 'usm_ighf_green',
           'usm_t_surf_wall', 'usm_t_surf_window', 'usm_t_surf_green',
           'usm_theta_10cm',
           'usm_t_wall', 'usm_t_window','usm_t_green',
           'usm_swc',
 /
```

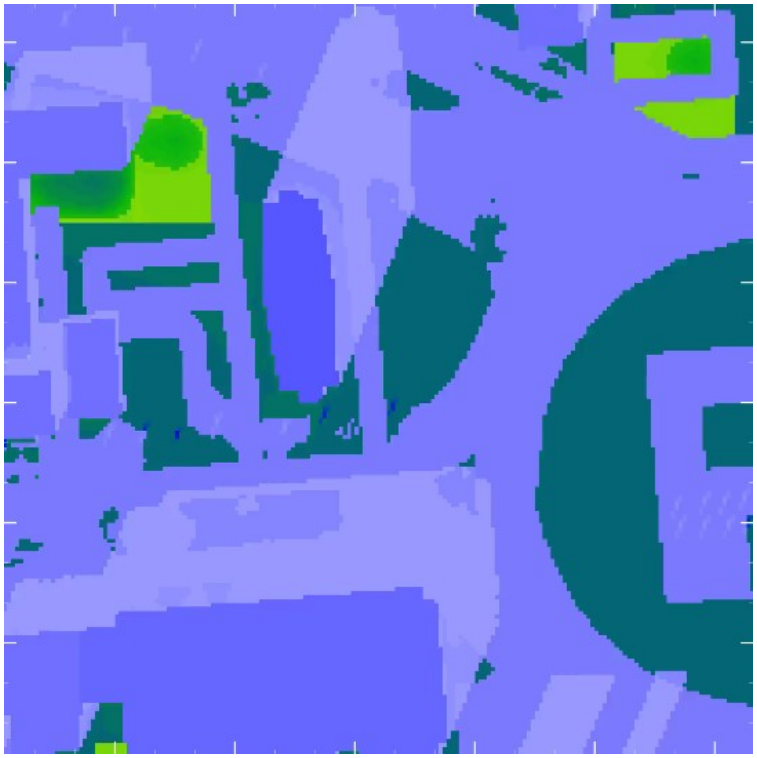
Limitations and upcoming improvements

- The BSM code still does not fully comply with PALM's coding standard and naming convention
- Some parts are deprecated and will be removed and or revised in near future (USM vs BSM naming convention)
- Slanted roofs are available (immersed boundary condition), but not well supported at the moment
- Green facade/roof substrate is always saturated
- More building types will be implemented (TABULA)
- Snowpack on buildings will be available soon
- Roof-top and building-integrated photovoltaic modules will be available soon
- Mid-term plan: a unified surface model

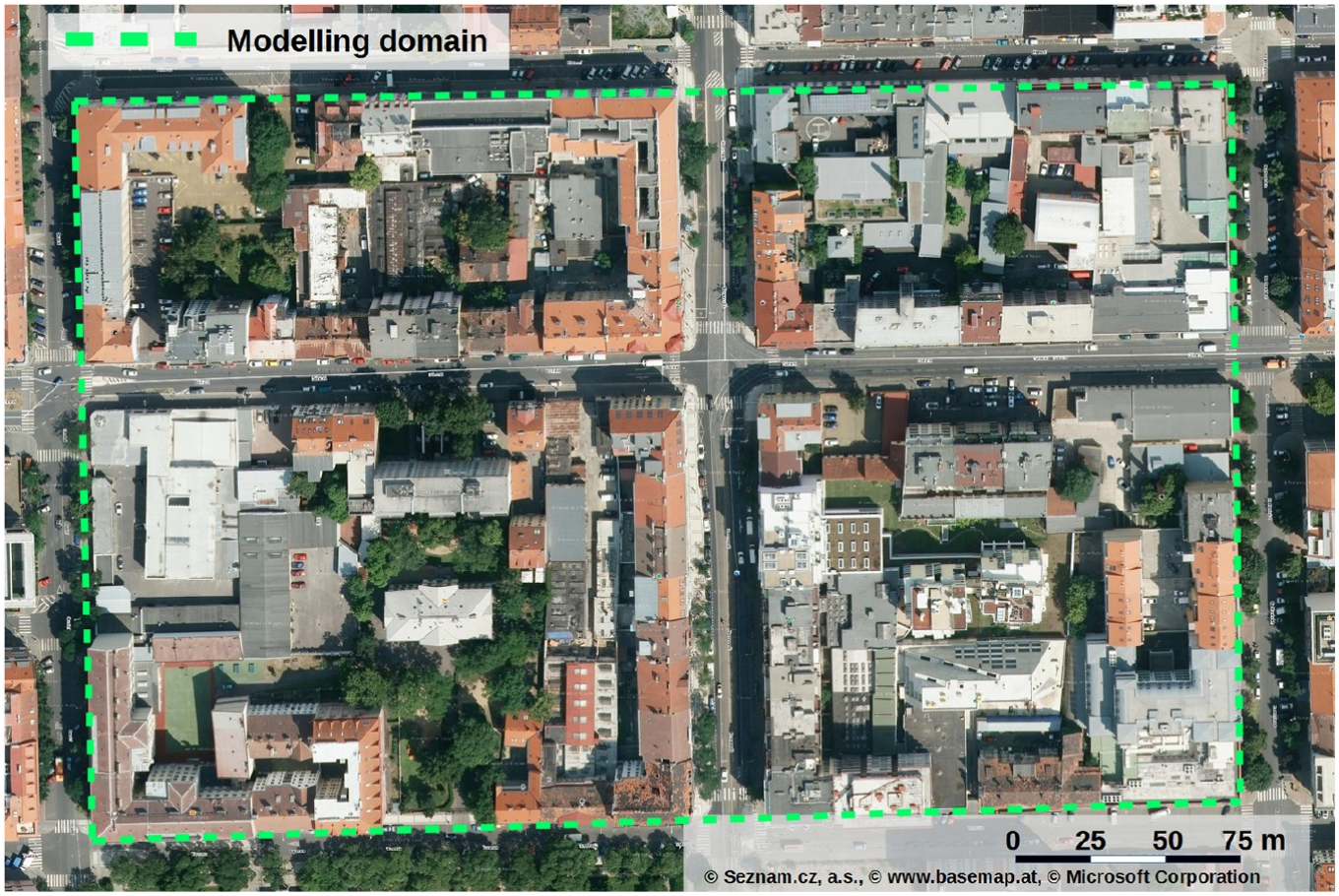
  → Working title: LAURA

  Land and Aquatic Unified model for Resolving surface-Atmosphere exchange

Surface temperature at Ernst-Reuter-Platz, Berlin



Surface temperatures in a city quarter of Prague



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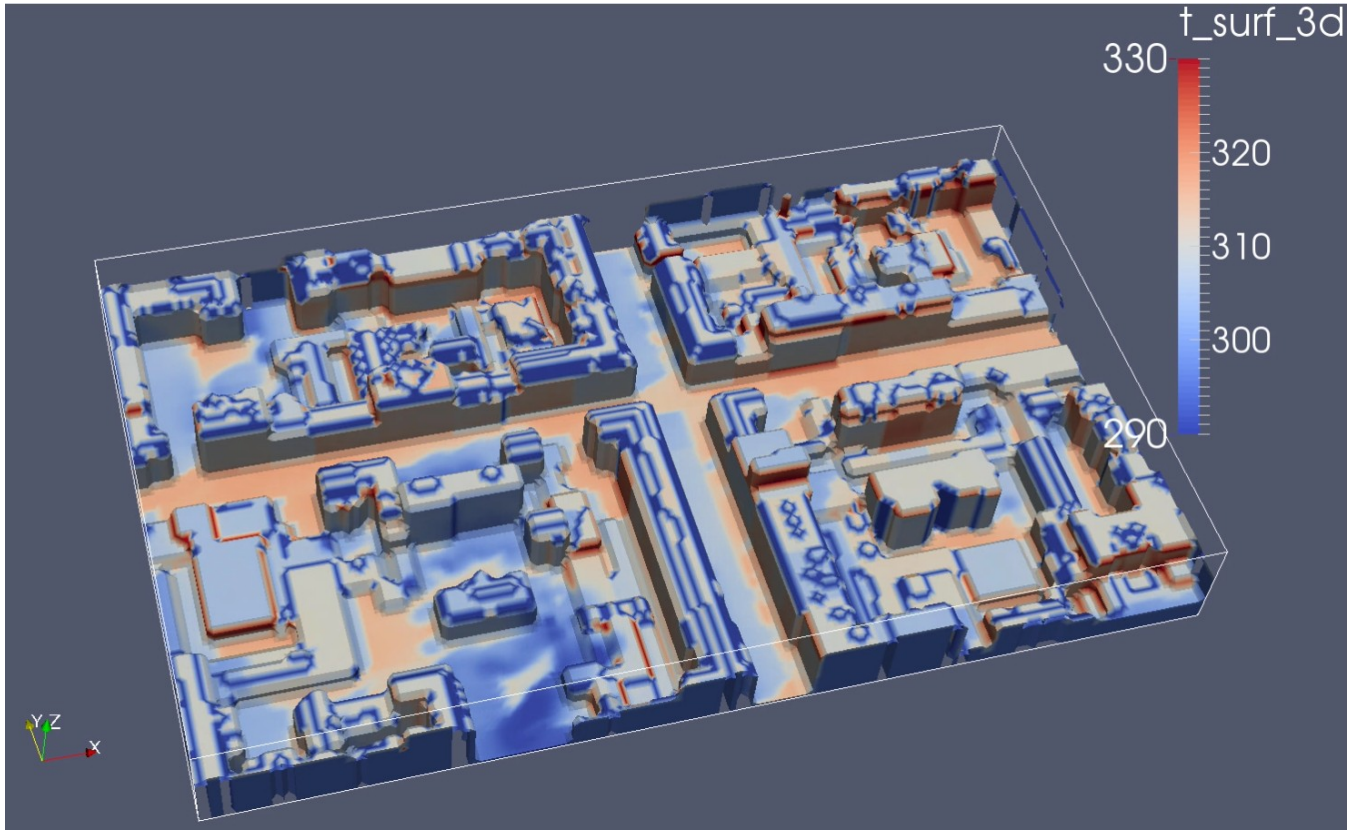

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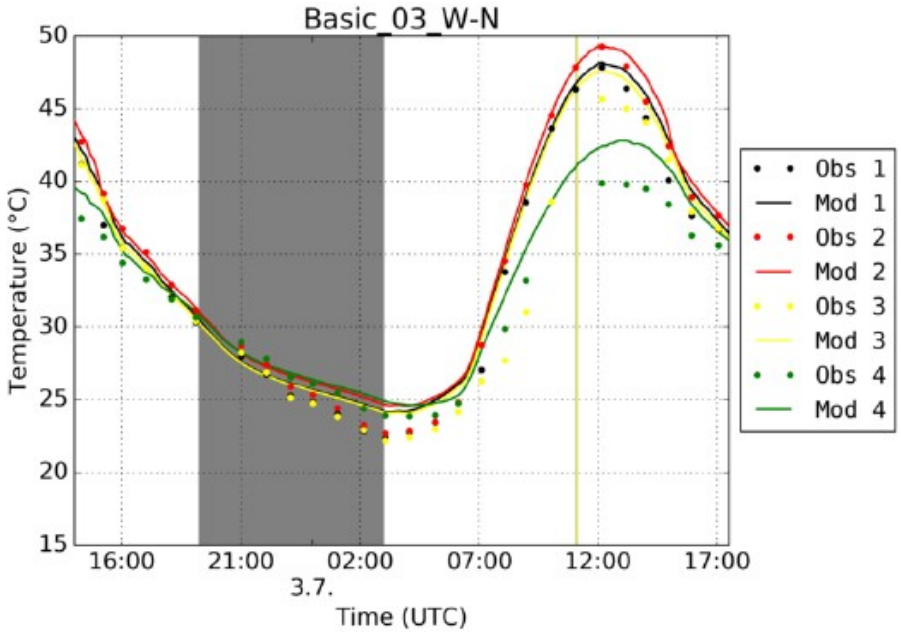

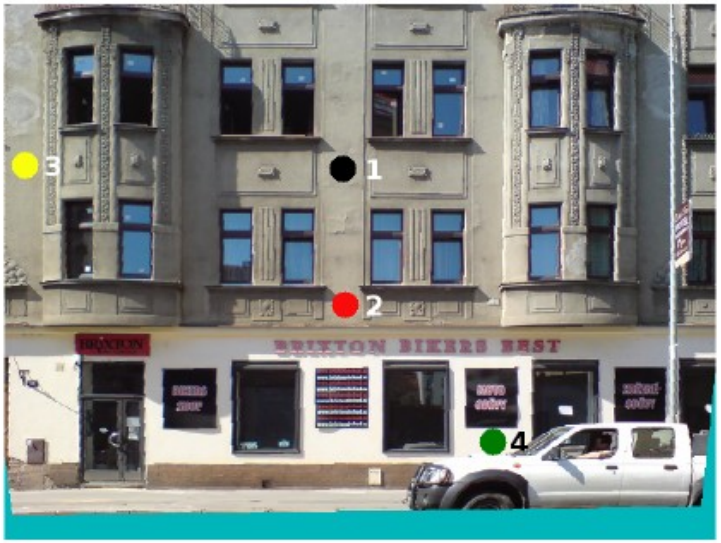

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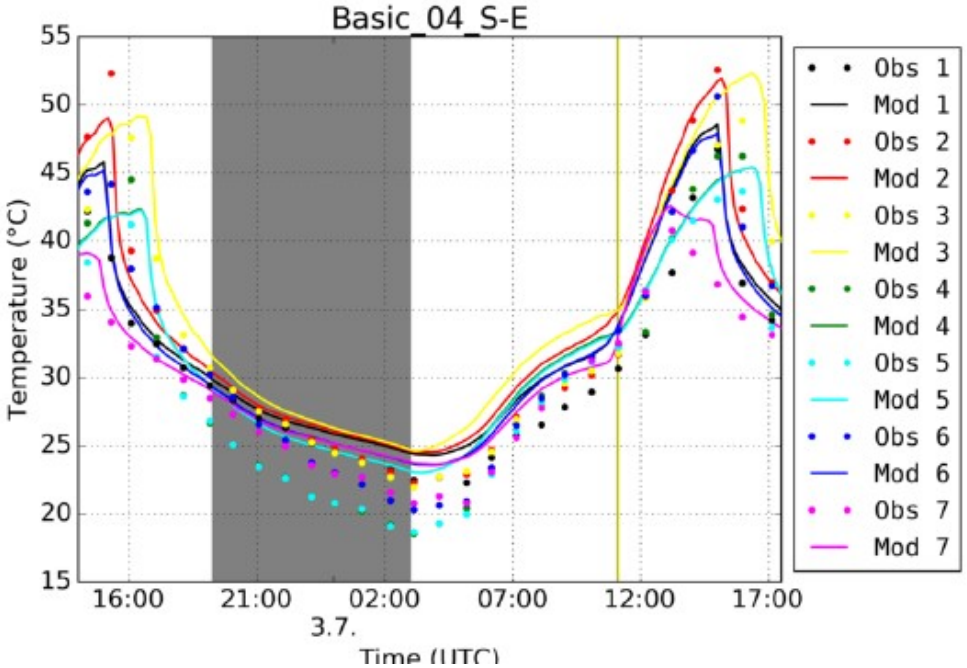

Indoor climate model (ICM)

- Based on an analytic solution of Fourier's equation
- Resistance model
- Time-stepping: Crank-Nicolson, dt = 1 h
- Output quantities: operative room temperature, energy demand for heating/cooling/lighting/ventilation, waste heat
- Uses the same information from the building database
- Coupling to BSM:
  - Inner wall/window/roof temperatures → indoor model
  - Near-facade temperature → indoor model
  - Inner wall heat flux → BSM

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  - Inner wall heat flux → BSM

Validation and first application:

- Pfafferott et al. (2021, 10.5194/gmd-14-3511-2021)
- Maronga et al. (2022, 10.1175/JAMC-D-21-0216.1)

Behavior of operative indoor temperature during a heat wave

Maronga et al. (2022, 10.1175/JAMC-D-21-0216.1)

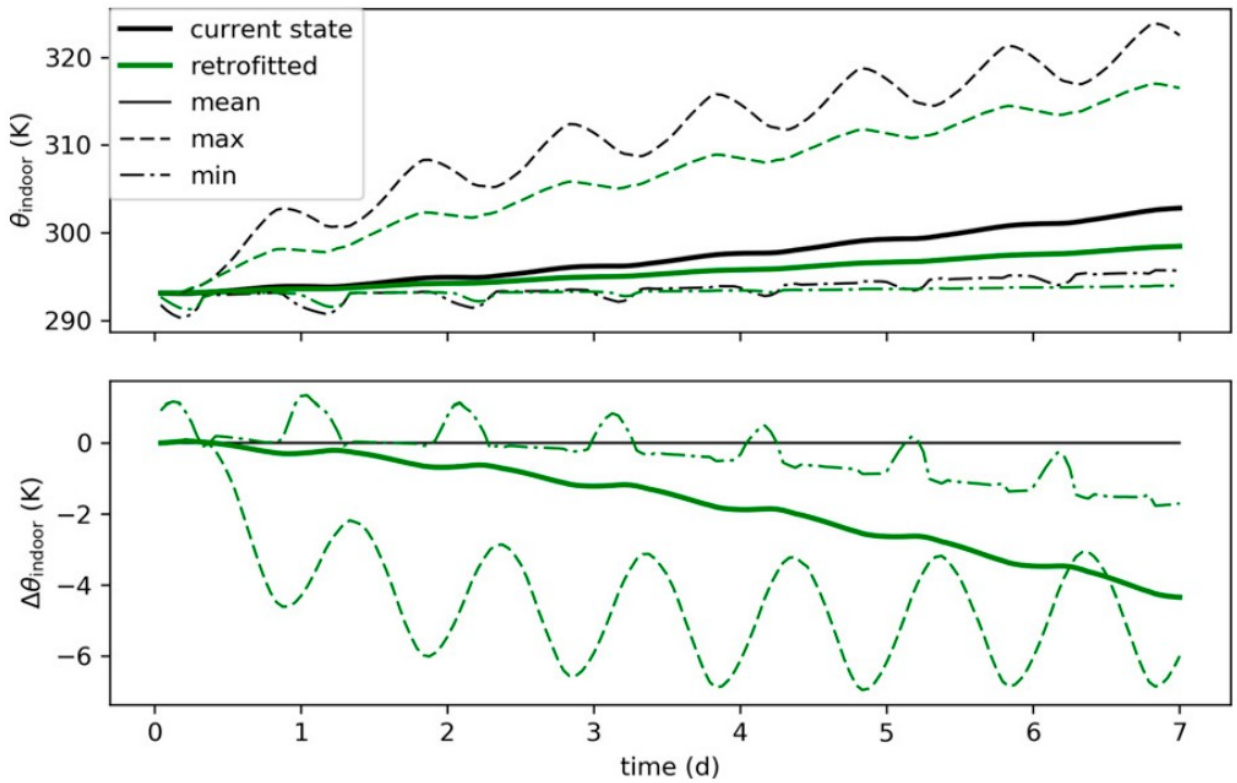

FIG. 6. Time series of (top) the indoor temperature and (bottom) the differences between the retrofitting and baseline runs. Shown are mean (domain averaged) values as well as maximum and minimum values. Note that indoor temperatures were averaged over each vertical column and thus the minimum and maximum temperatures indicate horizontal variations.

Usage I: Input parameters

- The ICM is activated via NAMELIST:

```
&indoor_parameters

   initial_indoor_temperature = 293.0,

 /
```

- Building types have to be given in static driver file

Usage II: Output quantities

- Output quantities

```
&runtime_parameters

   data_output = im_t_indoor_mean,
                 im_hf_roof,
                 im_hf_roof_waste,
                 im_hf_wall_win,
                 im_hf_wall_win_waste,
 /
```